

```
In [5]: import pandas as pd          #Loading a dataset(csv file):          Step 2: Load dataset
df=pd.read_csv('C:/Users/hp/Desktop/financial_sample_data.csv')
df.head()          #It view first row.
```

Out[5]:

	Date	Company	Ticker	Sector	Open Price	Close Price	Volume	Market Cap (in B)	PE Ratio	EPS
0	2025-04-01	Alpha Corp	ALPC	Technology	150.25	155.10	1200000	120.5	25.3	6.12
1	2025-04-01	Beta Inc	BETI	Healthcare	95.40	93.75	980000	45.8	18.6	5.04
2	2025-04-01	Gamma Ltd	GMLT	Finance	210.60	215.00	1540000	78.2	20.1	10.70
3	2025-04-01	Delta Energy	DLEN	Energy	45.75	47.90	2100000	30.3	12.9	3.71
4	2025-04-01	Sigma Retail	SGRT	Consumer Goods	60.10	58.50	750000	25.9	15.2	3.85

```
In [6]: df.info()          #Summary of data:          Step 3: Basic Data Overview
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  5 non-null     object
1   Company               5 non-null     object
2   Ticker                5 non-null     object
3   Sector                5 non-null     object
4   Open Price            5 non-null     float64
5   Close Price           5 non-null     float64
6   Volume                5 non-null     int64
7   Market Cap (in B)     5 non-null     float64
8   PE Ratio              5 non-null     float64
9   EPS                   5 non-null     float64
dtypes: float64(5), int64(1), object(4)
memory usage: 532.0+ bytes
```

```
In [7]: df.isnull().sum()          #Checks for missing values:          Step 4: Data Cleaning
```

```
Out[7]: Date                0
Company                  0
Ticker                  0
Sector                  0
Open Price              0
Close Price             0
Volume                  0
Market Cap (in B)       0
PE Ratio                0
EPS                    0
dtype: int64
```

```
In [12]: df.duplicated().sum()          #Step 4: Data Cleaning
```

Out[12]: np.int64(0)

In [13]: df.describe()

Out[13]:

	Open Price	Close Price	Volume	Market Cap (in B)	PE Ratio	EPS
count	5.000000	5.000000	5.000000e+00	5.000000	5.000000	5.000000
mean	112.420000	114.050000	1.314000e+06	60.140000	18.420000	5.884000
std	68.085191	70.266849	5.271432e+05	39.49029	4.77043	2.864669
min	45.750000	47.900000	7.500000e+05	25.900000	12.900000	3.710000
25%	60.100000	58.500000	9.800000e+05	30.300000	15.200000	3.850000
50%	95.400000	93.750000	1.200000e+06	45.800000	18.600000	5.040000
75%	150.250000	155.100000	1.540000e+06	78.200000	20.100000	6.120000
max	210.600000	215.000000	2.100000e+06	120.500000	25.300000	10.700000

In [17]: df['Price Change'] = df['Close Price'] - df['Open Price'] *#it added as new col*

In [21]: df.head() *#it typed again to check column 'price change' is added or not.*

Out[21]:

	Date	Company	Ticker	Sector	Open Price	Close Price	Volume	Market Cap (in B)	PE Ratio	EPS
0	2025-04-01	Alpha Corp	ALPC	Technology	150.25	155.10	1200000	120.5	25.3	6.12
1	2025-04-01	Beta Inc	BETI	Healthcare	95.40	93.75	980000	45.8	18.6	5.04
2	2025-04-01	Gamma Ltd	GMLT	Finance	210.60	215.00	1540000	78.2	20.1	10.70
3	2025-04-01	Delta Energy	DLEN	Energy	45.75	47.90	2100000	30.3	12.9	3.71
4	2025-04-01	Sigma Retail	SGRT	Consumer Goods	60.10	58.50	750000	25.9	15.2	3.85

In [20]: df['Percent Change'] = ((df['Price Change']) / df['Open Price']) * 100 *#Daily*

In [23]: df.groupby('Sector')['Close Price'].mean() *# Average Close Price by Sector: S*

Out[23]:

```
Sector
Consumer Goods    58.50
Energy            47.90
Finance           215.00
Healthcare        93.75
Technology        155.10
Name: Close Price, dtype: float64
```

In [24]: `df.groupby('Sector')['Volume'].sum()` *# Total Volume by Sector:*

Out[24]:

Sector	
Consumer Goods	750000
Energy	2100000
Finance	1540000
Healthcare	980000
Technology	1200000

Name: Volume, dtype: int64

In [26]: `df.loc[df['Market Cap (in B)'].idxmax()]` *# Company with Highest Market Cap:*

Out[26]:

Date	2025-04-01
Company	Alpha Corp
Ticker	ALPC
Sector	Technology
Open Price	150.25
Close Price	155.1
Volume	1200000
Market Cap (in B)	120.5
PE Ratio	25.3
EPS	6.12
Price Change	4.85
Percent Change	3.227953

Name: 0, dtype: object

In [27]: `df.sort_values('PE Ratio', ascending=False).head(3)` *#Step 7: Sorting
Top 3 Companies by PE R*

Out[27]:

	Date	Company	Ticker	Sector	Open Price	Close Price	Volume	Market Cap (in B)	PE Ratio	EPS
0	2025-04-01	Alpha Corp	ALPC	Technology	150.25	155.10	1200000	120.5	25.3	6.12
2	2025-04-01	Gamma Ltd	GMLT	Finance	210.60	215.00	1540000	78.2	20.1	10.70
1	2025-04-01	Beta Inc	BETI	Healthcare	95.40	93.75	980000	45.8	18.6	5.04

In [28]: `df.sort_values('EPS').head(3)` *# Lowest EPS*

Out[28]:

	Date	Company	Ticker	Sector	Open Price	Close Price	Volume	Market Cap (in B)	PE Ratio	EPS	Ch
3	2025-04-01	Delta Energy	DLEN	Energy	45.75	47.90	2100000	30.3	12.9	3.71	
4	2025-04-01	Sigma Retail	SGRT	Consumer Goods	60.10	58.50	750000	25.9	15.2	3.85	
1	2025-04-01	Beta Inc	BETI	Healthcare	95.40	93.75	980000	45.8	18.6	5.04	

```
In [29]: df.corr(numeric_only=True) #Step 8: Correlation Analysis
```

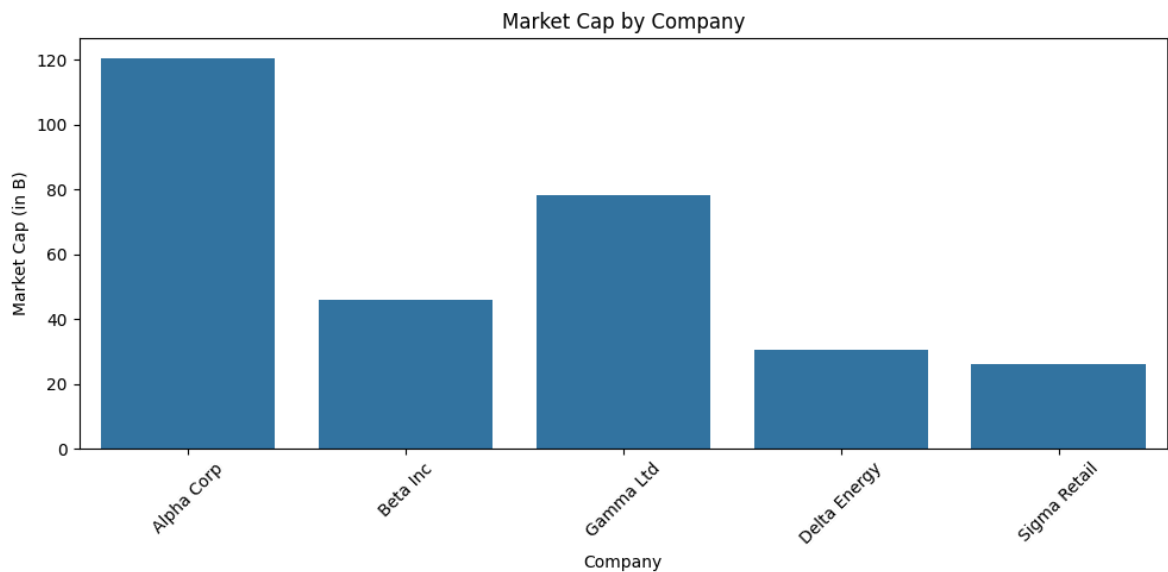
```
Out[29]:
```

	Open Price	Close Price	Volume	Market Cap (in B)	PE Ratio	EPS	Price Change	Percent Change
Open Price	1.000000	0.999465	0.004643	0.751430	0.737878	0.958113	0.682271	0.210458
Close Price	0.999465	1.000000	0.028695	0.762921	0.740137	0.956071	0.705811	0.240652
Volume	0.004643	0.028695	1.000000	-0.025854	-0.296994	0.129433	0.541008	0.874649
Market Cap (in B)	0.751430	0.762921	-0.025854	1.000000	0.952841	0.547708	0.778598	0.413798
PE Ratio	0.737878	0.740137	-0.296994	0.952841	1.000000	0.514291	0.562772	0.132193
EPS	0.958113	0.956071	0.129433	0.547708	0.514291	1.000000	0.619497	0.204894
Price Change	0.682271	0.705811	0.541008	0.778598	0.562772	0.619497	1.000000	0.821236
Percent Change	0.210458	0.240652	0.874649	0.413798	0.132193	0.204894	0.821236	1.000000

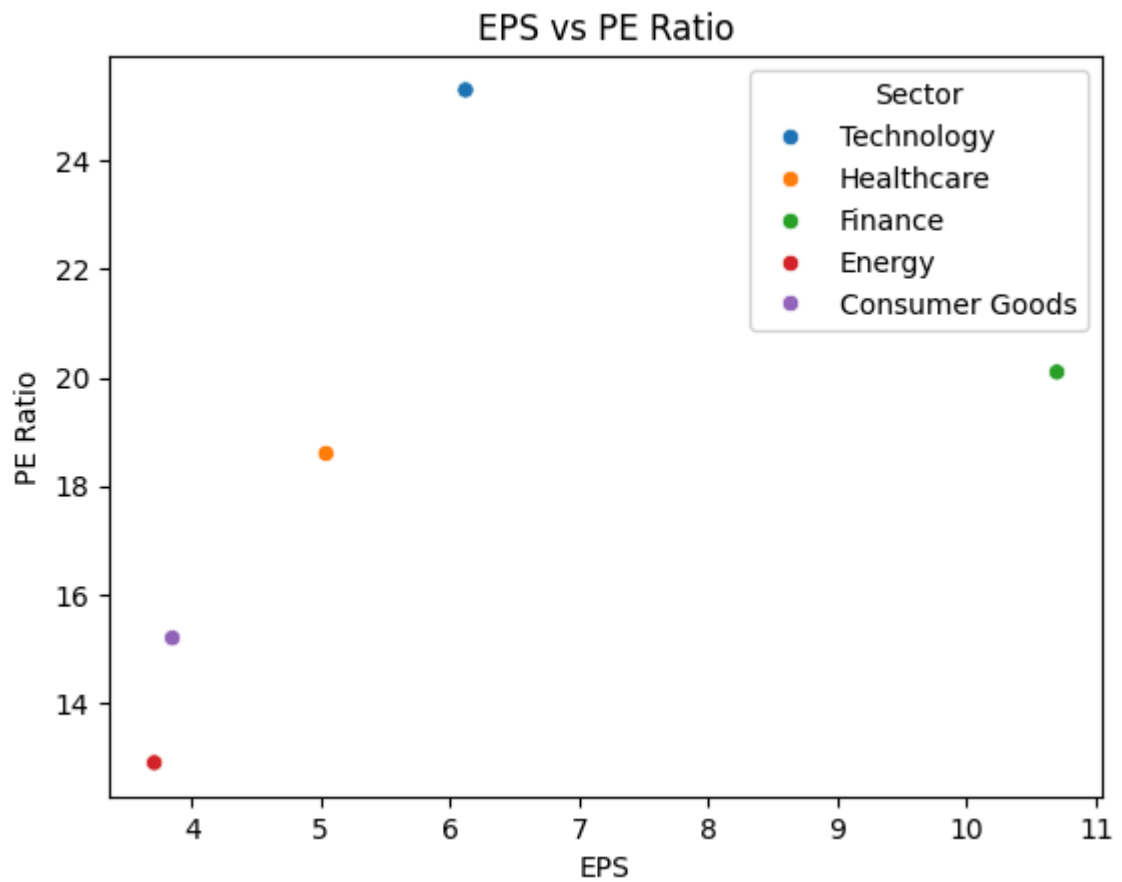


```
In [31]: import matplotlib.pyplot as plt #Step 9: Basic Visualization
import seaborn as sns
```

```
In [33]: # Bar chart - Market Cap by Company
plt.figure(figsize=(10, 5))
sns.barplot(data=df, x='Company', y='Market Cap (in B)')
plt.xticks(rotation=45)
plt.title('Market Cap by Company')
plt.tight_layout()
plt.show()
```



```
In [34]: # Scatter plot - EPS vs PE Ratio
sns.scatterplot(data=df, x='EPS', y='PE Ratio', hue='Sector')
plt.title('EPS vs PE Ratio')
plt.show()
```



```
In [37]: df.to_csv(r'C:\Users\hp\Desktop\cleaned_financial_data.csv', index=False) #S
```

```
In [ ]:
```